

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei200_off200GHmidFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei200_off200GHmidFF
```

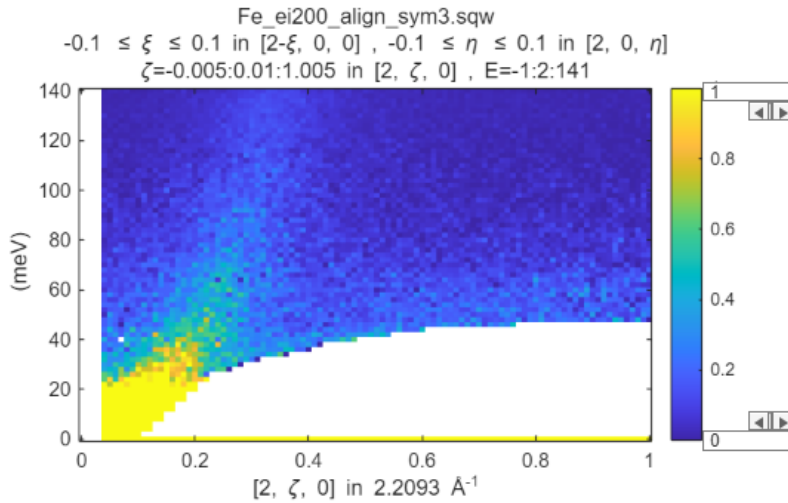
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in going finer
```

```
half_dE = 5; % half width of data binning
```

```
en_mm = cuts_list{1}.data.img_range(:,4);
```

```
en_mm(1) = max(min_Ei,en_mm(1));
```

```
%en_mm(1) = 100;
```

```
%en_mm(2) = 170;
```

```
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);
```

```
cut_dimensions = 1;
```

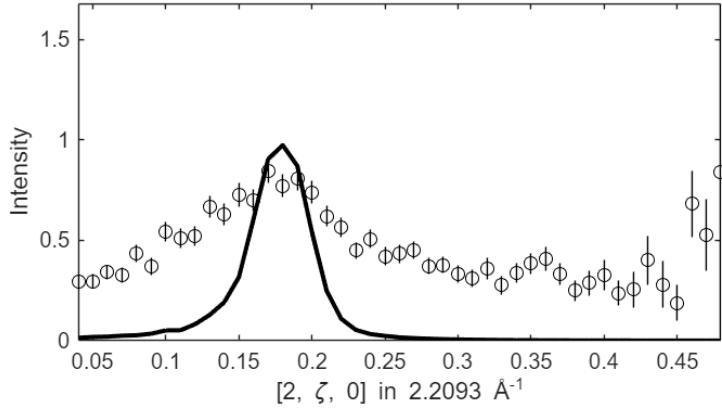
```
GH_FFCutsFits.(selectCuts) = cuts_list;
```

```
GH_FFCutsFits = fit_multicutsFF_along_direction(...
```

```
GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

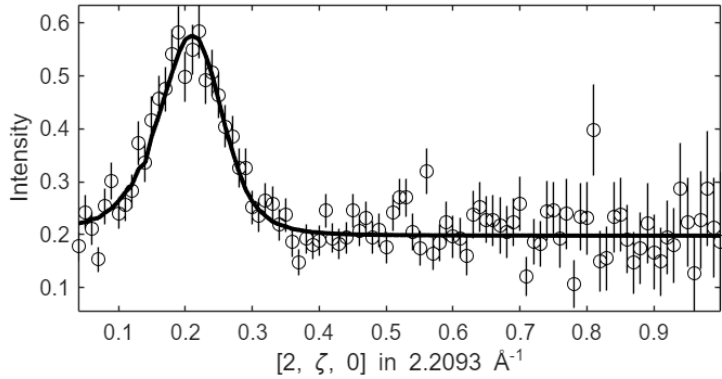
```
*****
**** En = 35±5 going to 135
*****
```

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $30 \leq E \leq 40$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

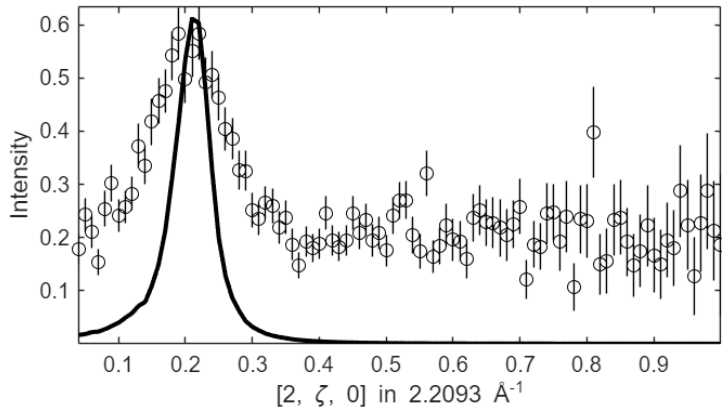


 **** En = 45±5 going to 135

Fe_ei200_align_sym3.sqw
 $S=0.89 \pm 0.04$; $J_0= 24 \pm 0.49$; $J_1= 0 \pm 0$; $\gamma=7.86132 \pm 0.401212$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $40 \leq E \leq 50$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



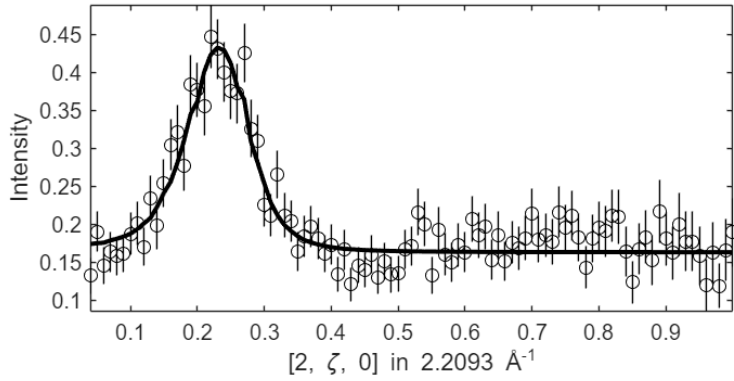
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $40 \leq E \leq 50$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



 **** En = 55±5 going to 135

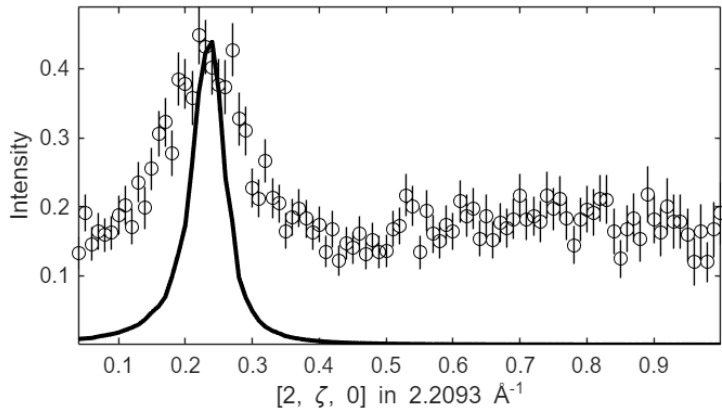
Fe_ei200_align_sym3.sqw

S=0.69±0.028; J0= 24± 0.5; J1= 0± 0; gamma=9.15344±0.398561
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq E \leq 100$ eV
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$



Fe_ei200_align_sym3.sqw

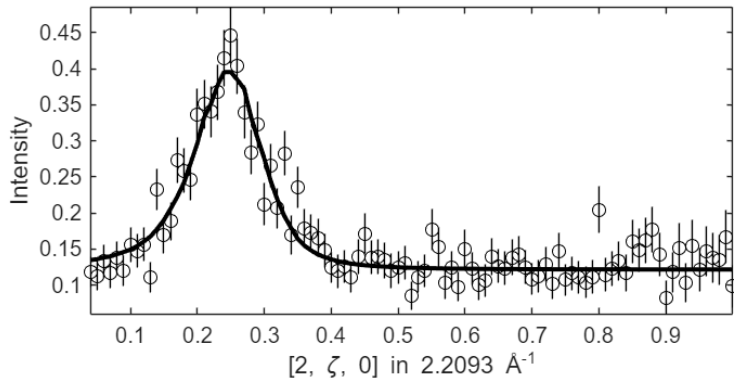
S=0.69±0.028; J0= 24± 0.5; J1= 0± 0; gamma=9.15344±0.398561
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq E \leq 100$ eV
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$



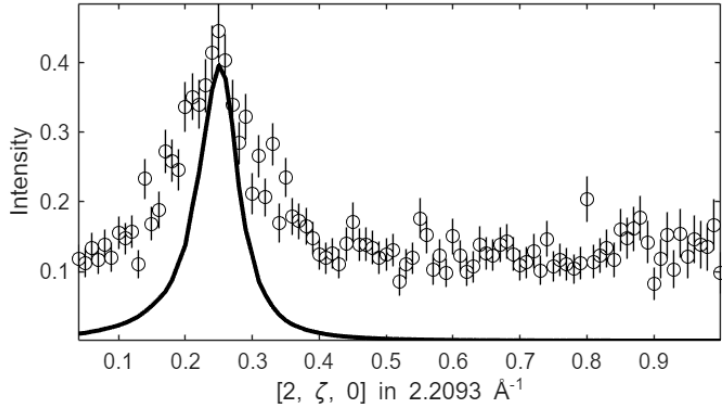
 **** En = 65±5 going to 135

Fe_ei200_align_sym3.sqw

S=0.93±0.035; J0= 26±0.49; J1= 0± 0; gamma=15.0241±0.558638
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E \leq 100$ eV
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

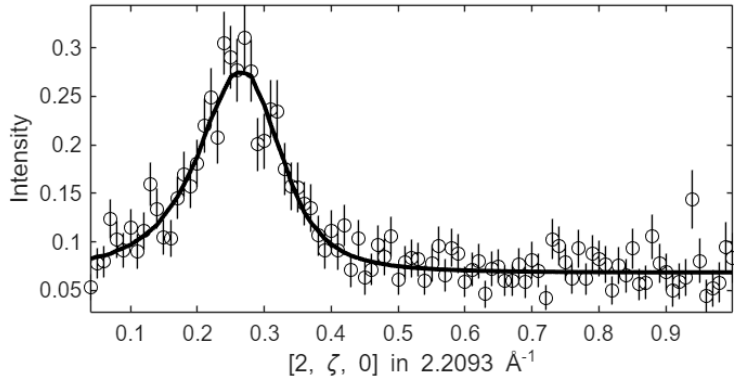


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E \leq 70$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

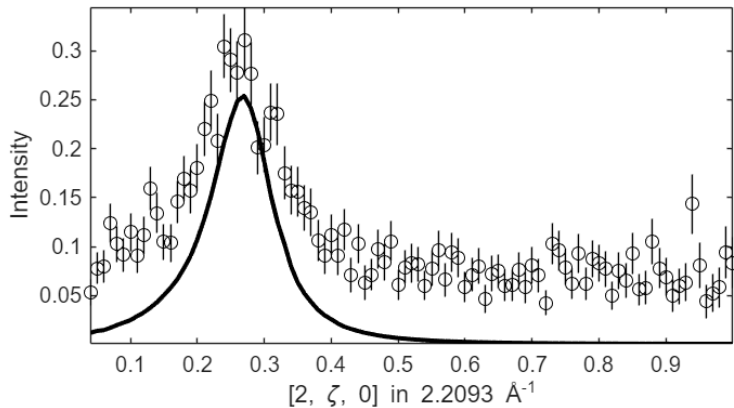


 **** En = 75±5 going to 135

Fe_ei200_align_sym3.sqw
 $S = 1.1 \pm 0.039$; $J_0 = 28 \pm 0.55$; $J_1 = 0 \pm 0$; $\gamma = 28.8058 \pm 0.926367$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $70 \leq E \leq 80$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



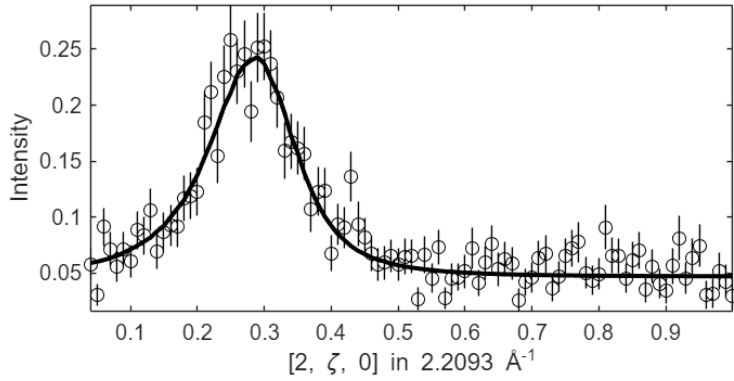
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $70 \leq E \leq 80$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



 **** En = 85±5 going to 135

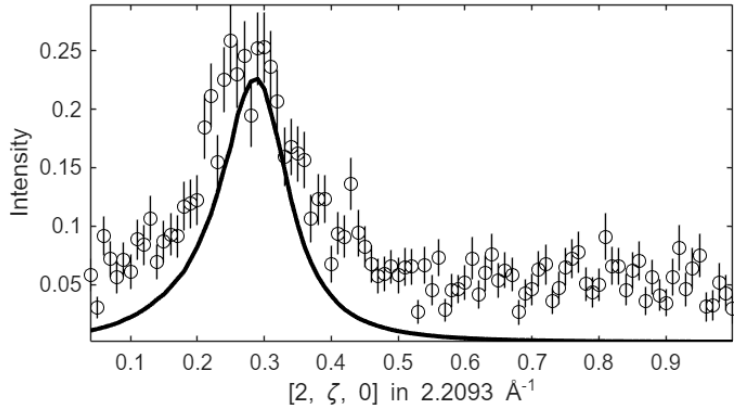
Fe_ei200_align_sym3.sqw

S= 1.2±0.11; J0= 28±0.88; J1= 0± 0; gamma=35.5106±5.47564
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$ eV
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$



Fe_ei200_align_sym3.sqw

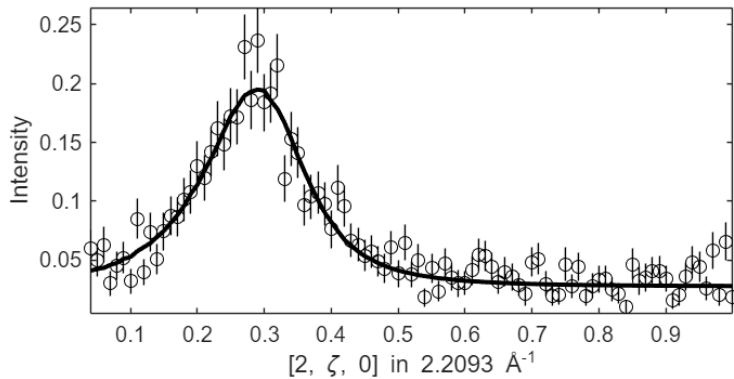
$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$ eV
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$



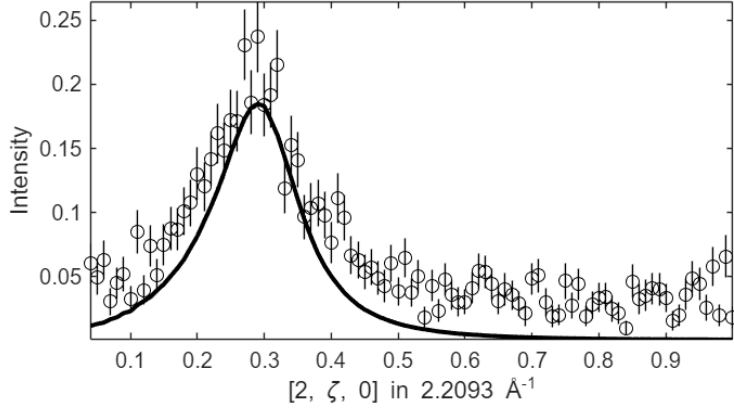
 **** En = 95±5 going to 135

Fe_ei200_align_sym3.sqw

S= 1.4±0.15; J0= 33± 1.4; J1= 0± 0; gamma=55.2178±9.16342
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $90 \leq E \leq 110$ eV
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

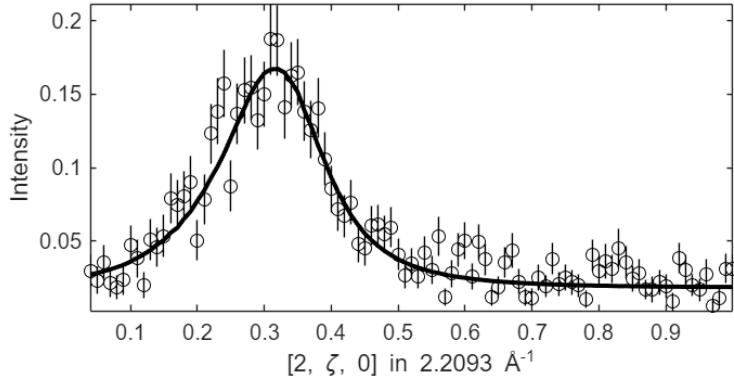


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $90 \leq E \leq 1$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

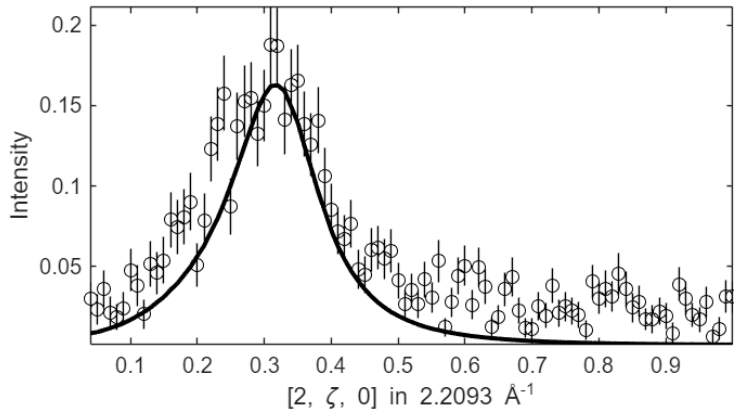


 **** En = 105±5 going to 135

Fe_ei200_align_sym3.sqw
 $S = 1.3 \pm 0.15$; $J_0 = 31 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 58.1893 \pm 9.63048$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $100 \leq E \leq$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

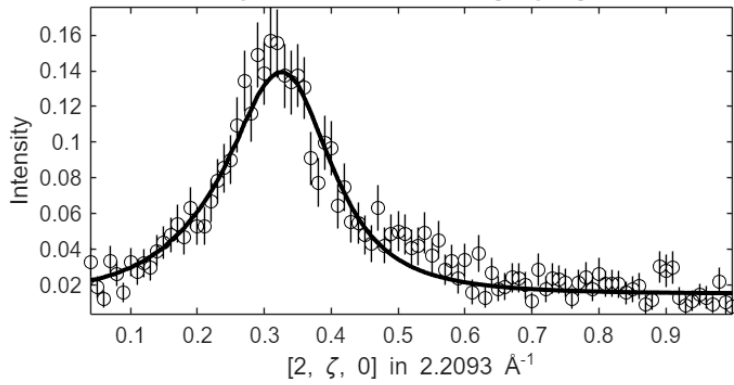


Fe_ei200_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $100 \leq E \leq$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

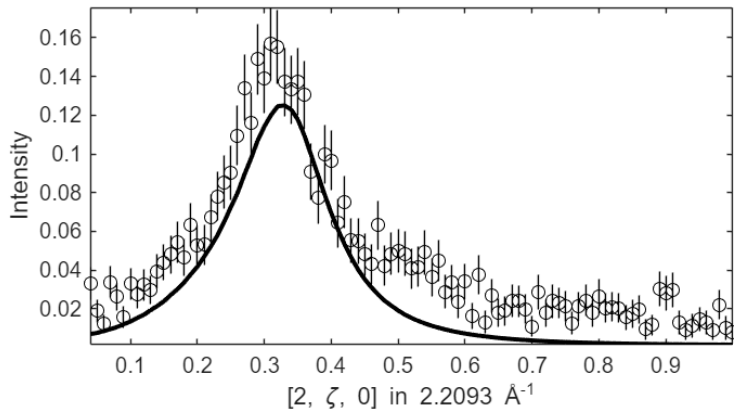


 **** En = 115±5 going to 135

Fe_ei200_align_sym3.sqw
 $S = 1.2 \pm 0.098$; $J_0 = 32 \pm 0.98$; $J_1 = 0 \pm 0$; $\gamma = 66.2904 \pm 8.3646$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $110 \leq E \leq 110$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

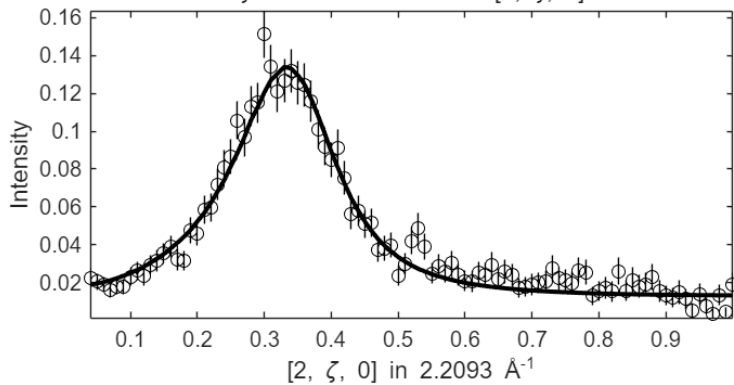


Fe_ei200_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $110 \leq E \leq 110$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

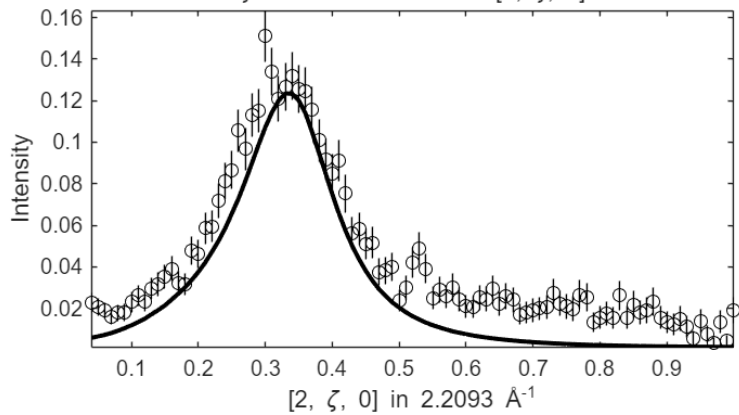


 **** En = 125±5 going to 135

Fe_ei200_align_sym3.sqw
 $S = 1.2 \pm 0.077$; $J_0 = 34 \pm 0.76$; $J_1 = 0 \pm 0$; $\gamma = 69.4227 \pm 6.45225$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq 120$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

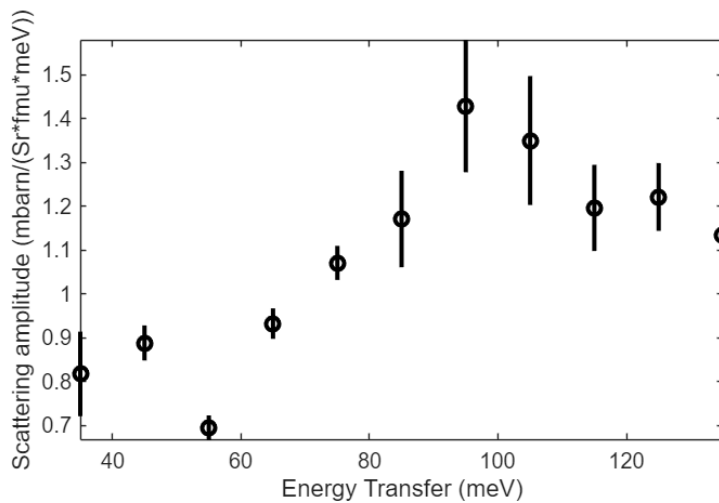
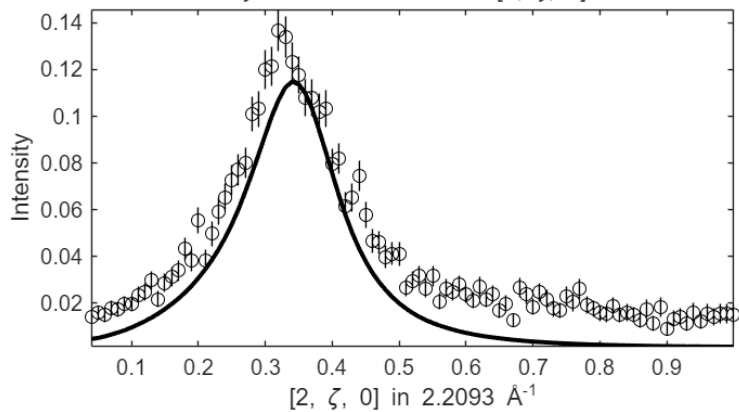


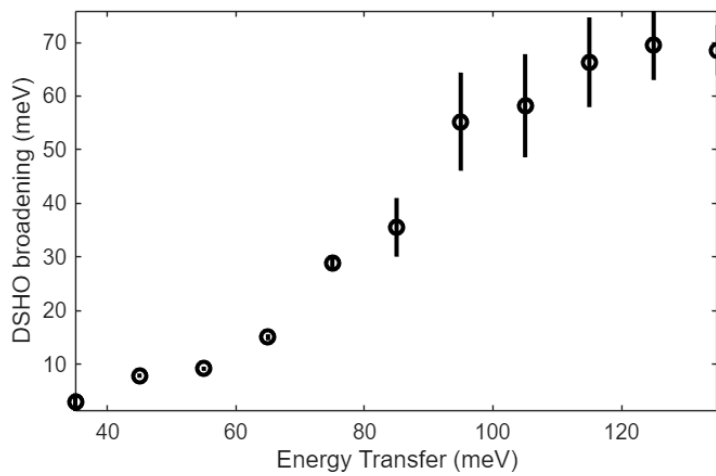
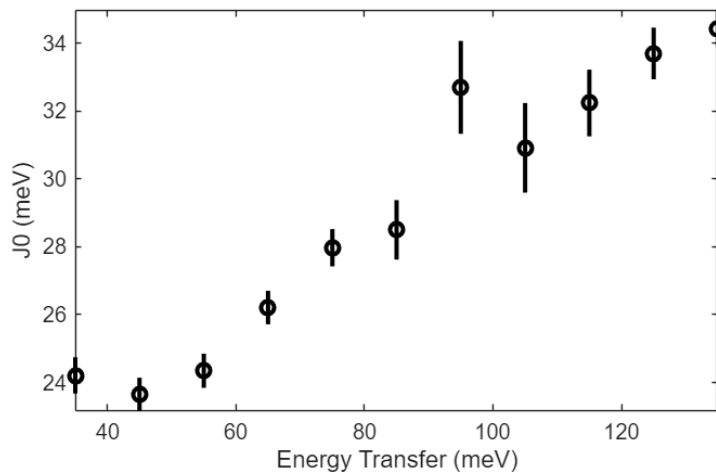
Fe_ei200_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$



**** En = 135 ± 5 going to 135

Fe_ei200_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $130 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$





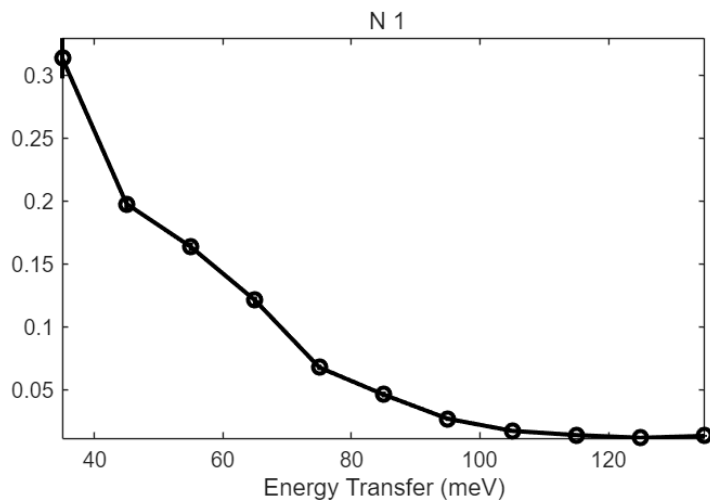
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

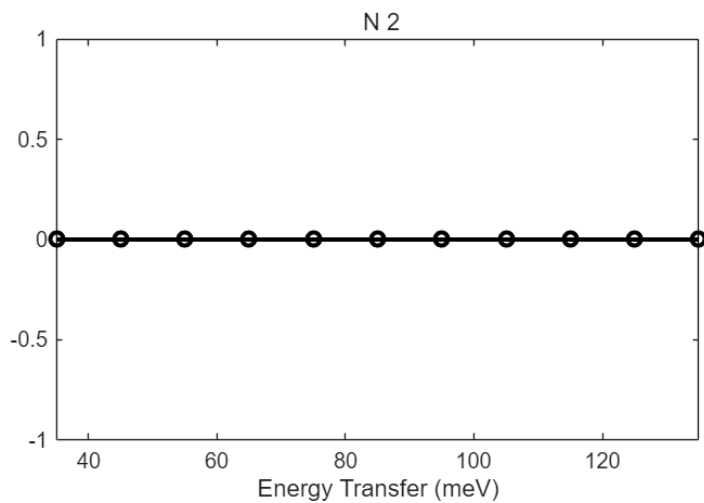
beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struct",-v7.3');
close all;
%end
```